Group 3 - Real-Time Object Detection

Objective:

Implement **real-time object detection** using **YOLO** on a **live video feed** and compare its performance with **R-CNN-based models** in terms of inference speed and efficiency on the same hardware. This experiment will help analyze the feasibility of deploying object detection models in real-time applications.

Task Breakdown:

1. Setting Up Live Video Feed

- Capture a real-time video feed using a **webcam, USB camera, or an IP camera**.
- Ensure the video feed has a stable frame rate and resolution (e.g., **640×480 or 1280×720**).
- Implement a pipeline for reading frames in real-time using **OpenCV**.

---

2. Implementing YOLO for Real-Time Object Detection

- Use a pre-trained **YOLOv5** or **YOLOv8** model.
- Load the model and optimize it for real-time inference:
    - Resize input frames to match YOLO's expected input dimensions.
    - Perform **frame-by-frame** inference.
    - Display detected objects in real-time with bounding boxes and confidence scores.
- Measure **frames per second (FPS)** for YOLO detection.

---

3. Implementing R-CNN-Based Model for Object Detection

- Use a **Faster R-CNN** model trained on the same dataset.
- Process each video frame using Faster R-CNN:
    - Convert the frame to the required input format.
    - Perform **region proposal-based object detection**.
    - Display detected objects with bounding boxes and confidence scores.
- Measure **FPS** for Faster R-CNN detection.

---

4. Performance Comparison

Evaluate and compare the two models using the following metrics:

A. Inference Speed (FPS)

- Measure the **frames per second (FPS)** for both YOLO and Faster R-CNN.
- Analyze the speed difference and its impact on real-time applications.

B. Detection Accuracy

- Compare the **bounding box quality and confidence scores**.
- Analyze detection delays and how they affect real-time usability.

C. Resource Utilization

- Measure **CPU/GPU usage and memory consumption** for both models.
- Discuss **hardware requirements** for real-time deployment.

---

5. Final Report & Analysis

Each group must submit a report including:

1. **FPS comparison** between YOLO and Faster R-CNN.
2. **Detection accuracy analysis** for both models.
3. **Resource utilization study** – CPU/GPU usage, memory consumption.
4. **Graphical analysis**:
    - FPS comparison chart.
    - Sample frames showing detection outputs.
5. **Discussion on trade-offs**:
    - **When to use YOLO vs. R-CNN for real-time applications?**
    - **What hardware is required for deploying these models in real-world settings?**

Deliverables

✓ **Code Implementation**: Real-time object detection using YOLO and Faster R-CNN.
✓ **Performance Metrics**: FPS, detection accuracy, and resource utilization.

✓ **Comparison Report**: Findings and analysis of real-time feasibility.

✓ **Live Demo** : Presenting real-time object detection performance.